

AURACLOCK S3

MECHANICAL & ELECTRICAL DIMENSIONS SPECIFICATION

Project Type:	IoT Focus Hub & Mechatronic System	Date:	May 18, 2026
Lead Engineer:	Ibrahim Abdull (CEO Ibrahim)	Platform Targeting:	Hack Club Fallout Submission
Enclosure Core:	Onshape Parametric Assembly	PCB Environment:	EasyEDA Pro Turnkey PCBA

This document details the exact technical dimensions, real-world measurements, mechanical tolerances, and hardware parameters for the **AuraClock S3** focus hub assembly. To comply with the strict guidelines of the Hack Club Fallout evaluation process, all structural parts, internal tracks, mechatronic interfaces, and fastening hardware (screws and threaded inserts) have been modeled natively from scratch. No third-party CAD files or downloaded fastener profiles are permitted.

1. MASTER ASSEMBLY & ENCLOSURE ENCLOSURE BOX DATA

The physical profile follows a 3:1 panoramic aspect ratio optimized for flat desktop environments. The outer structure features two split components: the Primary Rear Enclosure Box shell and the Front Retention Bezel Frame. Clearances are computed using non-overlapping fitment loops to guarantee effortless sliding paths on typical FDM printers.

MECHANICAL PART COMPONENT	DIMENSION AXIS PARAMETER	TARGET SIZE	ENGINEERING CLEARANCE / TOLERANCE CONSTRAINTS
Primary Rear Box Shell	X-Axis (Total Width)	180.00 mm	Baseline bounding datum plane edge.
	Y-Axis (Total Height)	60.00 mm	Baseline bounding datum plane height.
	Z-Axis (Total Depth)	45.00 mm	Uniform structural shell thickness of 2.50 mm.
Front Window Acrylic Tray	Width Cutout	180.30 mm	Includes a +0.30 mm offset relative to panel lens size.
	Height Cutout	60.30 mm	Includes a +0.30 mm offset relative to panel lens size.
	Z-Depth Pocket	2.20 mm	Hosts 2.00 mm smoked acrylic sheet (+0.20 mm step clearance).
Internal Moving Shutter Blind	Width Scale	110.00 mm	Ensures a full opaque field mask over the utility screen array.
	Height Scale	52.00 mm	Provides structural clearance inside the 55.00 mm inner void.
	Thickness	2.50 mm	Constructed to minimize mass and sliding resistance inertia.
Ceiling/Floor Guide Channels	Track Width Open	3.10 mm	Provides a +0.30 mm alignment path to prevent slide binding.
	Track Total Depth	3.00 mm	Extruded straight into upper and lower case limits.
Main Exterior Corners	Fillet Sweep Radius	6.00 mm	Applied along vertical edge lines to mimic reference panel profile.

CRITICAL MANUFACTURING NOTE: THE 0.3MM CLEARANCE LAW

FDM 3D printing layers tend to swell on horizontal paths. To prevent the mechanical slide plate from seizing inside the case, all moving boundaries utilize an explicit mechanical tolerance gap of **0.30mm**.

2. TURNKEY PCB SMT FOOTPRINT & LAYOUT BOUNDARIES

The electrical architecture relies on a custom multi-layer circuit board optimized for automated SMT factory production. The physical perimeter shape has been derived from an Onshape vector sketch to line up perfectly with internal screw pillars.

PCB FUNCTIONAL LAYER METRIC	PHYSICAL PARAMETER	TARGET DIMENSION	SOURCING CATALOG / SMT LAYOUT DESIGN CONSTRAINTS
Turnkey Circuit Board Frame	Total Copper Width	150.00 mm	Fits within internal 175.00 mm inner enclosure limit.
	Total Copper Height	48.00 mm	Sits comfortably below lower ventilation duct openings.
	FR4 Substrate Depth	1.60 mm	Standard multi-layer operational core specification.
ESP32 RF Antenna Zone	Keep-out Width	15.00 mm	Zero trace copper routing permitted across all layers.
	Keep-out Height	10.00 mm	Module trace edge overhangs external frame boundary by 1.50 mm.
Power Distribution Rails	Trace Copper Width	1.00 mm	Sized for 12V inputs, 5V motor grids, and 3.3V logic planes.
Signal Communication Paths	Trace Copper Width	0.25 mm	Applies to I2C lines (SDA/SCL) and I2S digital audio channels.
PCB Mechanical Fastening Holes	Hole Core Diameter	3.20 mm	Provides standard pass-through clearance for M3 machine screws.
	Top-Left Location	(5.0, 5.0) mm	Absolute coordinates mapped relative to top-left board origin.
	Top-Right Location	(145.0, 5.0) mm	Absolute coordinates mapped relative to top-left board origin.
	Bottom-Right Location	(145.0, 43.0) mm	Absolute coordinates mapped relative to top-left board origin.

3. CUSTOM-MODELED HARDWARE FASTENERS (SCREWS & POCKETS)

To prevent project rejection, standard engineering fasteners have been fully sketched and modeled inside a dedicated Part Studio. Case connections rely on brass threads melted directly into integrated internal pillars.

FASTENER COMPONENT PROFILE	DIMENSIONAL THREAD ATTRIBUTE	TARGET MEASUREMENT	ONSHAPE PARAMETRIC FEATURE INSTRUCTIONS
Primary Structural Case Screw (M3 Model Profile)	Thread Outer Diameter	3.00 mm	Fully modeled solid cylinder body path.
	Thread Core Pitch	0.50 mm	Derived using the uniform helix spiral parameter tool.
	Total Stem Length	12.00 mm	Passes through the 2.50 mm front frame straight to internal bosses.
	Cap Socket Diameter	5.50 mm	ISO 4762 Socket Head specification replicated 1:1.
	Cap Socket Height	3.00 mm	Hexagonal drive socket cut down 1.50 mm deep.
Brass Heat-Set Insert Pocket (Case Pillars)	Outer Boss Wall Width	6.00 mm	Cylindrical standoff columns printed onto rear frame interior.
	Pocket Entrance Hole	4.20 mm	Sized narrow to let plastic melt cleanly around brass knurling.
	Pocket Total Depth	5.50 mm	Ensures a flush fit for a 5.00 mm long insert block.
	Internal Thread Metric	M3 x 0.5	Receives the custom-modeled structural case bolts.
OLED Internal Retention Fastener (M2.5 Model Profile)	Thread Outer Diameter	2.50 mm	Secures display module tabs directly to internal bezel tabs.
	Thread Core Pitch	0.45 mm	Modeled spiral helix configuration pattern.
	Total Stem Length	6.00 mm	Short form factor to avoid pushing through front face.

4. CUSTOM MECHATRONIC RACK-AND-PINION GEAR METRICS

The physical shutter assembly uses a custom-modeled gear transmission layer that translates the rotation of the SG90 servo into flat linear travel. All dimensions are derived strictly from standard Module 1.0 gear design formulas.

Formula Reference: Pitch Diameter (D) = $M \times N$ | Outside Diameter (D_a) = $D + 2M$

KINEMATIC SYSTEM ELEMENT	GEAR PARAMETER NAME	TARGET NUMERIC METRIC	ANALYTICAL REFERENCE FORMULA
Servo Drive Pinion Gear (Spur Component)	Fundamental Module (M)	1.00	Sized for durable tooth structures on standard 3D printers.
	Number of Teeth (N)	12	Ensures quick travel speeds within tight space constraints.
	Pitch Circle Diameter	12.00 mm	$D = 1.0 \times 12$
	Outside Tip Diameter	14.00 mm	$D_a = 12.0 + (2 \times 1.0)$
	Root Base Diameter	9.50 mm	Includes a standard dedendum clearance constant.
	Face Depth Width	6.00 mm	Extruded along the gear axis for reliable mechanical mesh.
Linear Mask Shutter Rack Track	Linear Tooth-to-Tooth Pitch	3.142 mm	$P = \pi \times M$
	Total Linear Length	94.25 mm	Contains 30 individual teeth paths carved horizontally.
	Working Cut Depth	2.25 mm	Determines the height from the root to the tooth tip.
	Total Mask Travel Distance	60.00 mm	Achieved via a 180-degree servo sweep to blind the display.
Interaxial Mounting Axis	Center-to-Pitch Line Gap	7.00 mm	The physical distance between the servo axis and the rack base plane.

End of Engineering Specification // AuraClock S3 Blueprint Framework V1.0 // Ibrahim Abdull